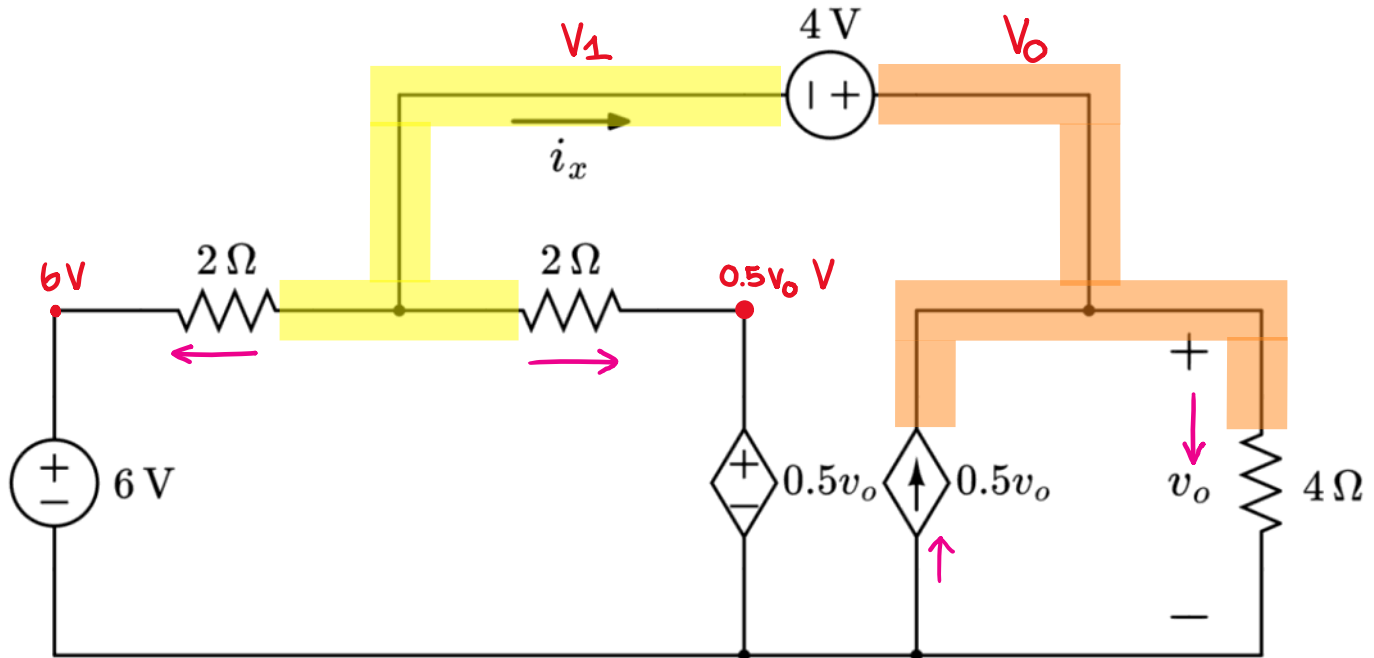


Question One: Node Voltage Analysis

For the circuit shown below, use node voltage analysis, find the following:

- a) v_o
- b) i_x

Step One: Write supernode constraint equation



$$V_0 - V_1 = 4$$

Step Two: KCL at supernode (V_1 and V_0) Left to Right

$$\frac{V_1 - 6}{2} + \frac{V_1 - 0.5v_0}{2} - 0.5v_0 + \frac{V_0}{4} = 0$$

$$\frac{V_1}{2} - \frac{6}{2} + \frac{V_1}{2} - \frac{0.5V_0}{2} - \underline{0.5v_0} + \frac{V_0}{4} = 0$$

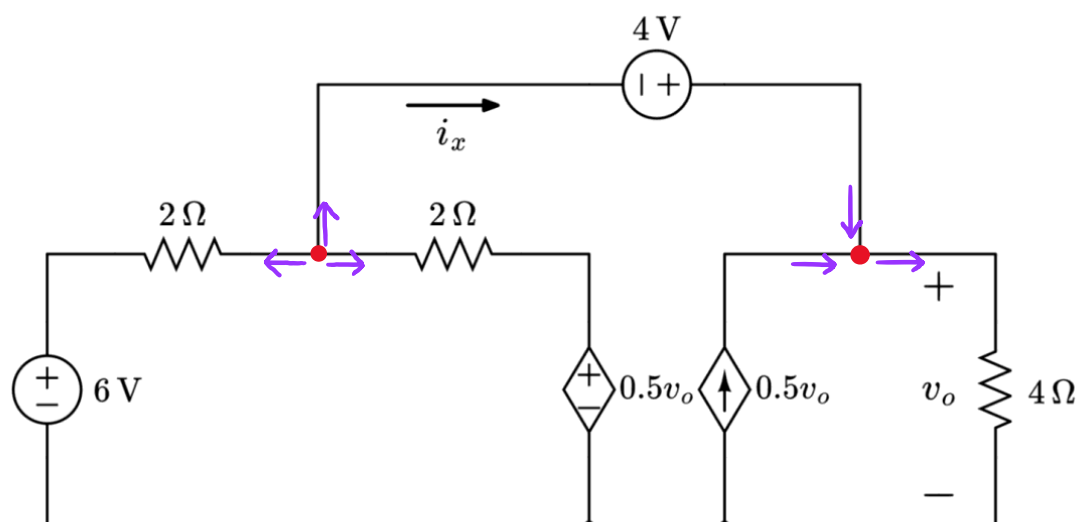
$\quad + \frac{6}{2} \qquad \qquad \frac{0.5}{2} = \frac{1}{4} \qquad \qquad + \frac{6}{2}$

$$V_1 - \frac{1}{2}V_0 = 3 \longrightarrow -\frac{1}{2}V_0 + V_1 = 3$$

Step Three: Formulate, solve 2×2 matrix

$$\begin{bmatrix} 1 & -1 \\ -\frac{1}{2} & 1 \end{bmatrix} \begin{bmatrix} V_0 \\ V_1 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \end{bmatrix} \longrightarrow \begin{bmatrix} V_0 \\ V_1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -\frac{1}{2} & 1 \end{bmatrix}^{-1} \begin{bmatrix} 4 \\ 3 \end{bmatrix} \quad \begin{bmatrix} V_0 \\ V_1 \end{bmatrix} = \begin{bmatrix} 14 \\ 10 \end{bmatrix} \text{ V}$$

Step Four: Solve for i_x



KCL at V_0 KCL at V_1 would also work

$$-0.5v_o - i_x + \frac{v_o}{4} = 0 \quad v_o = 14V$$

$$-i_x - 3.5 = 0$$

$$i_x = -3.5A$$

Self-Grading Rubric	
Clearly name and label each node	+1 point
Correct KCL equation at the supernode	+4 points
Correct supernode constraint equation	+1 point
Correct answer for v_o and the second node voltage	+2 points
Correct KCL equation involving i_x	+1 point
Correct answer for i_x	+1 point

Suggested discussion problems:

- Discussion 4: 1-6
- Discussion 7: 1, 3

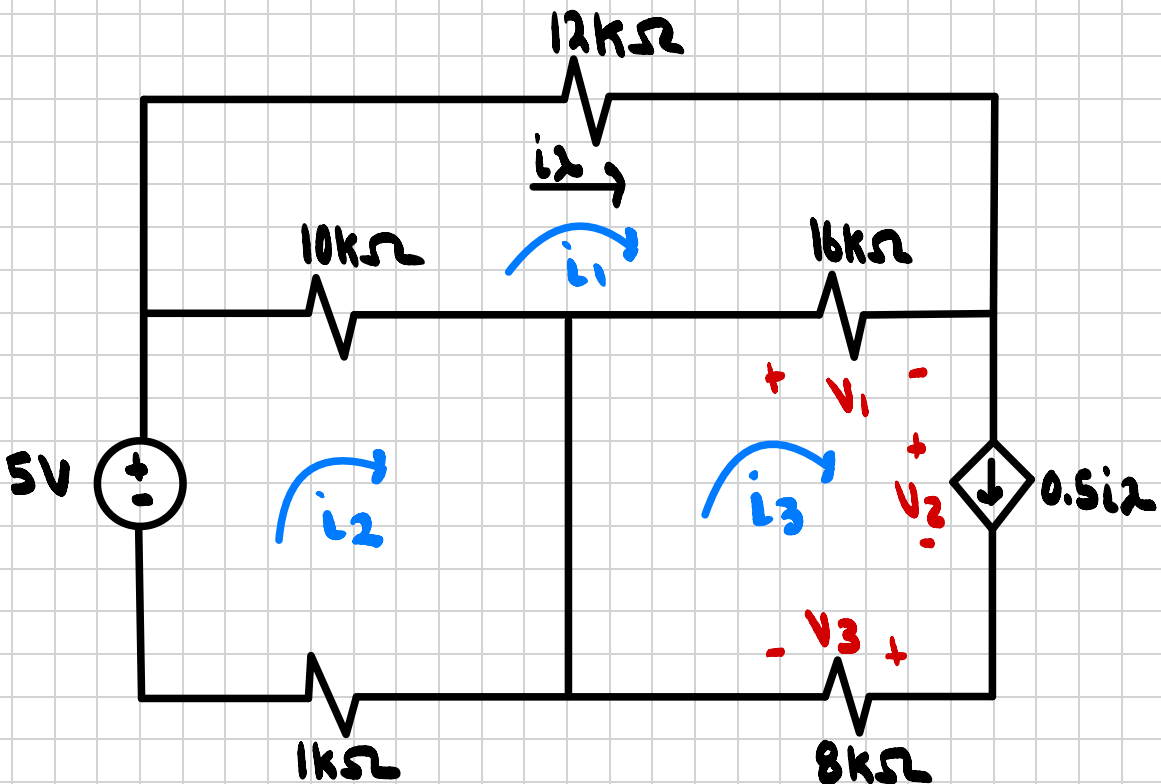
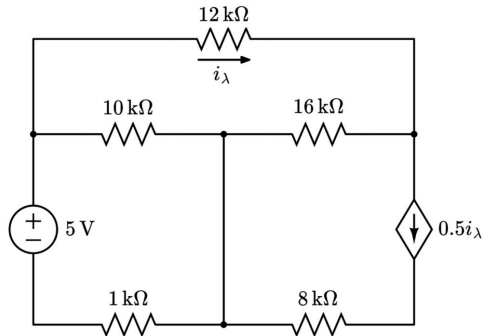
Question Two: Mesh Current Analysis

For the circuit shown below, use mesh current analysis, label and solve for each mesh current.

Finally, calculate the power of the current-dependent current source and state whether the power is absorbed or delivered.

Using node voltage analysis will result in zero credit earned, regardless of the final answer.

Suggested time limit: 20 minutes



Step #1 Constraint

$$i_1 = i_2$$

$$i_3 = 0.5i_2 \rightarrow i_3 = 0.5i_1 \rightarrow 0.5i_1 - i_3 = 0$$

Step #2 i_1 Mesh

$$12000i_1 + 16000(i_1 - i_3) + 10000(i_1 - i_2) = 0$$

$$12000i_1 + 16000i_1 - 16000i_3 + 10000i_1 - 10000i_2 = 0$$

$$38000i_1 - 10000i_2 - 16000i_3 = 0$$

Step #3 i_2 Mesh

$$-5 + 10000(i_2 - i_1) + 1000i_2 = 0$$

$$10000i_2 - 10000i_1 + 1000i_2 = 5$$

$$-10000i_1 + 11000i_2 = 5$$

Step #4 Matrix

$$\begin{bmatrix} 0.5 & 0 & -1 \\ 38000 & -10000 & -16000 \\ -10000 & 11000 & 0 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 5 \end{bmatrix}$$

$$i_1 = \frac{1}{4600} A$$

$$i_2 = \frac{3}{4600} \text{ A}$$

$$i_3 = \frac{1}{9200} \text{ A}$$

Step # 5 Power

$$V_1 + V_2 + V_3 = 0$$

$$16000(i_3 - i_1) + V_2 + 8000i_3 = 0$$

$$V_2 = -16000 \left(\frac{1}{9200} - \frac{1}{4600} \right) - 8000 \left(\frac{1}{9200} \right) = 0$$

$$V_2 = 0.8695652146 \text{ V}$$

$$P = V_2 \cdot 0.5 i_2$$

$$P = (0.8695652146)(0.5) \left(\frac{1}{4600} \right)$$

$$P = 94.5179642 \text{ W Absorb}$$

Grading Rubric

Constraint - 2 Point

Mesh @ i_1 - 2 Points

Mesh @ i_2 - 2 Points

Find i_1, i_2, i_3 - 1 Point

KVL @ Bottom Right Loop - 1 Point

Power Calculation - 1 Point

Absorb / Delivered - 1 Point

Discussion #5 Problems To Practice

Problem # 1

Problem # 2

Problem # 3

Problem # 5

Problem # 7

Question Three: Thévenin and Norton Equivalent Circuits with Maximum Power Transfer

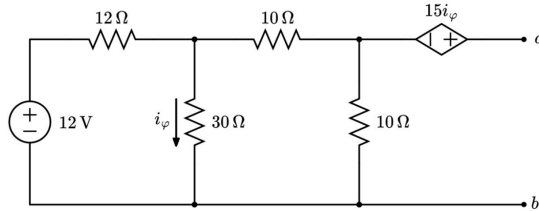
For the circuit shown below:

- Find the Thévenin equivalent circuit with respect to the open terminal pair ab and clearly illustrate the equivalent circuit with every value labeled.
- If a load is connected to points a and b , what is the maximum power that can be delivered to a load resistor R_L ?
- Suppose that a $22\ \Omega$ resistor is connected to points a and b . Find the voltage, current and power of the $22\ \Omega$ resistor and state whether the resistor is absorbing or delivering power.

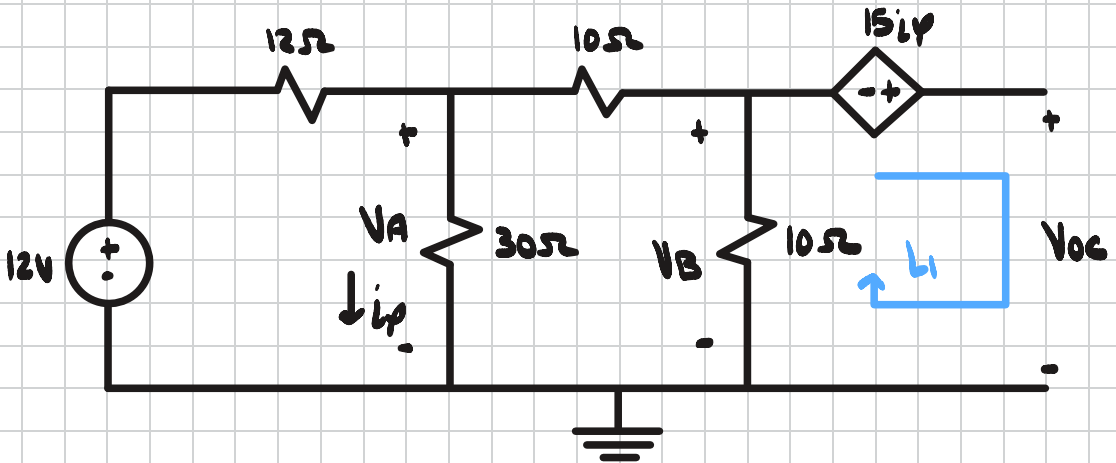
Node Voltage, Mesh Current,

KVL/KCL are not needed.

Suggested time limit: 20 minutes



Voc Method



Step #1 Constraint

$$i_\phi = \frac{V_A - 0}{30} \rightarrow i_\phi = \frac{V_A}{30}$$

Step #2 VA Nodal

$$\left(\frac{V_A - 12}{12} + \frac{V_A - 0}{30} + \frac{V_A - V_B}{10} = 0 \right) 60$$

$$5V_A - 60 + 2V_A + 6V_A - 6V_B = 0$$

$$13V_A - 6V_B = 60$$

Step #3 VB Nodal

$$\left(\frac{V_B - 0}{10} + \frac{V_B - V_A}{10} = 0 \right) 10$$

$$V_B + V_B - V_A = 0$$

$$-V_A + 2V_B = 0$$

Step #4 Matrix To Solve

$$\begin{bmatrix} 13 & -6 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} V_A \\ V_B \end{bmatrix} = \begin{bmatrix} 60 \\ 0 \end{bmatrix}$$

$$V_A = 6V$$

$$V_B = 3V$$

Step #5 Find ip

$$i_p = \frac{6}{30} = 0.2A$$

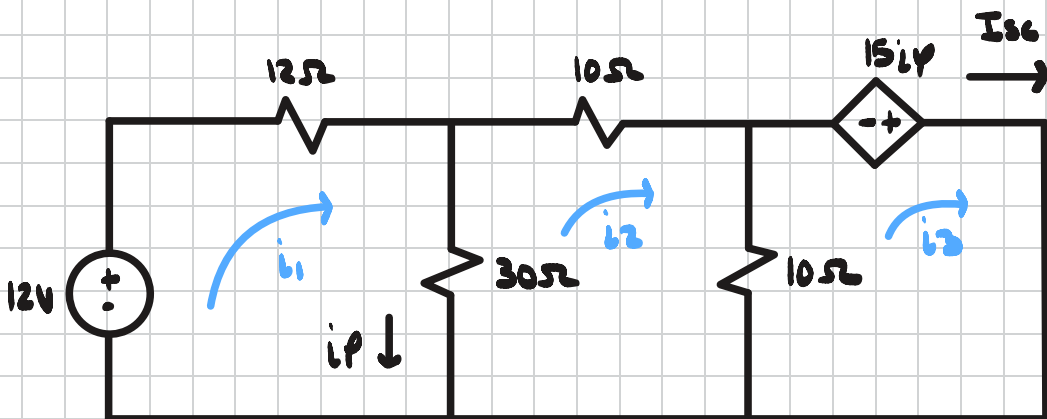
Step # 6 KVL @ L_1

$$-V_B - 15i_P + V_{oc} = 0$$

$$-3 - 15(0.2) + V_{oc} = 0$$

$$V_{oc} = 6V = V_{TH}$$

IsC Method



Step # 1 Constraint

$$i_P = i_1 - i_2$$

$$i_3 = I_{sc}$$

Step # 2 L_1 Mesh

$$-12 + 12i_1 + 30(i_1 - i_2) = 0$$

$$12i_1 + 30i_1 - 30i_2 = 12$$

$$42i_1 - 30i_2 = 12$$

Step #3 i₂ Mesh

$$30(i_2 - i_1) + 10i_2 + 10(i_2 - i_3) = 0$$

$$30i_2 - 30i_1 + 10i_2 + 10i_2 - 10i_3 = 0$$

$$-30i_1 + 50i_2 - 10i_3 = 0$$

Step #4 i₃ Mesh

$$10(i_3 - i_2) - 15i_1 = 0$$

$$10i_3 - 10i_2 - 15(i_1 - i_2) = 0$$

$$10i_3 - 10i_2 - 15i_1 + 15i_2 = 0$$

$$-15i_1 + 5i_2 + 10i_3 = 0$$

Step #5 Matrix To Solve

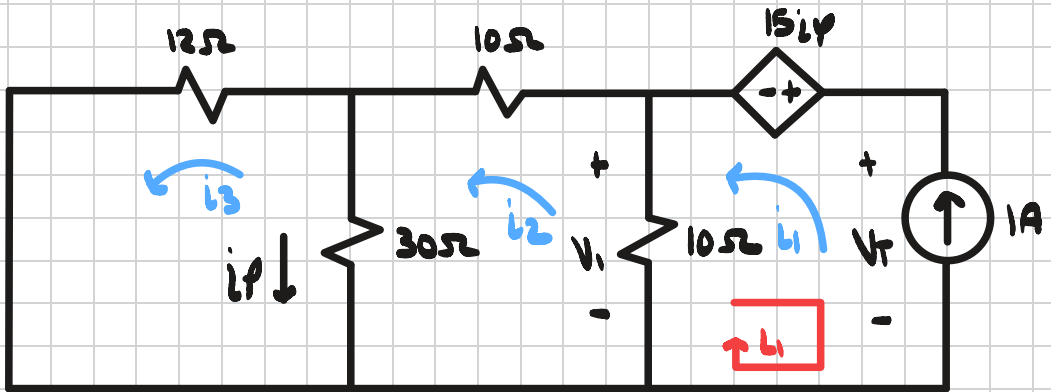
$$\begin{bmatrix} 42 & -30 & 0 \\ -30 & 50 & -10 \\ -15 & 5 & 10 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 12 \\ 0 \\ 0 \end{bmatrix}$$

$$i_1 = \frac{11}{16} A$$

$$i_2 = \frac{9}{16} A$$

$$i_3 = \frac{3}{4} A = I_{sc} = I_N$$

Test Source Method



Step #1 Constraint

$$i_1 = i_2 - i_3$$

$$i_1 = 1A = I_T$$

Step #2 i_2 Mesh

$$10(i_2 - i_1) + 10i_2 + 30(i_2 - i_3) = 0$$

$$10(i_2 - 1) + 10i_2 + 30(i_2 - i_3) = 0$$

$$10i_2 - 10 + 10i_2 + 30i_2 - 30i_3 = 0$$

$$50i_2 - 30i_3 = 10$$

Step #3 i_3 Mesh

$$30(i_3 - i_2) + 12i_3 = 0$$

$$30i_3 - 30i_2 + 12i_3 = 0$$

$$-30i_2 + 42i_3 = 0$$

Step # 4 Matrix To Solve

$$\begin{bmatrix} 50 & -30 \\ -30 & 42 \end{bmatrix} \begin{bmatrix} i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 0 \end{bmatrix}$$

$$i_2 = \frac{7}{20} A$$

$$i_3 = -\frac{1}{4} A$$

Step # 5 KVL L1

$$-V_1 \cdot 15i_1 + V_T = 0$$

$$-10(i_1 - i_2) - 15(i_2 - i_3) + V_T = 0$$

$$-10i_1 + 10i_2 - 15i_2 + 15i_3 + V_T = 0$$

$$-10i_1 - 5i_2 + 15i_3 = 0$$

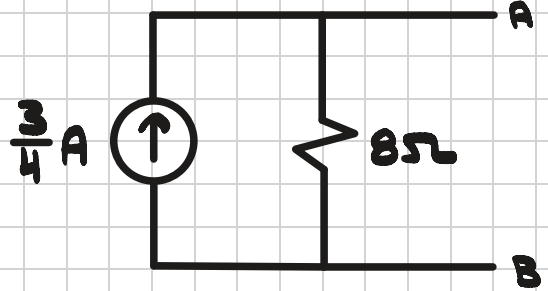
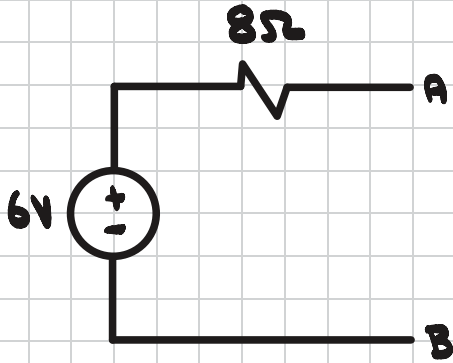
$$-10(1) - 5\left(\frac{7}{20}\right) + 15\left(-\frac{1}{4}\right) + V_T = 0$$

$$-8 + V_T = 0$$

$$V_T = 8V$$

$$\frac{V_T}{I_T} = \frac{8}{1} = 8\Omega = R_{TH}$$

Equivalent Circuit



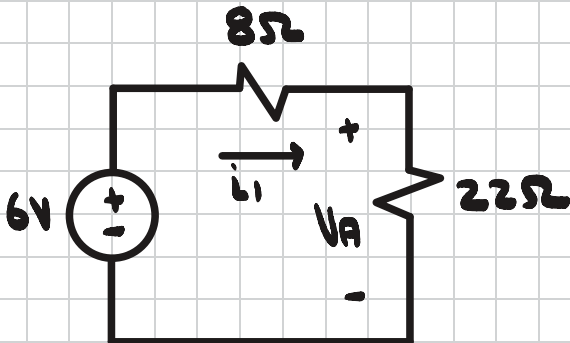
Max Power

$$P_{MAX} = \frac{(V_{TH})^2}{4R_{TH}}$$

$$P_{MAX} = \frac{(6)^2}{4(8)}$$

$$P_{MAX} = 1.125W \text{ Absorb}$$

Voltage/Current/Power For $R_L = 22\Omega$



$$V_A = 6 \left(\frac{22}{8+22} \right) = 4.4V$$

$$i_1 = \frac{6}{22+8} = \frac{6}{30} = 0.2A$$

$$P = VI = (4.4)(0.2) = 0.88W \text{ Absorb}$$

Grading Rubric

V_{TH} - 2 Points

I_N - 2 Points

R_{TH} - 2 Points

Equivalent Circuit - 1 Point

Max Power - 2 Points

Voltage/Current/Power For $R_L = 22\Omega$ - 1 Point

Discussion #5 Problems To Practice

Problem # 1

Problem # 2

Problem # 3

Problem # 4

Problem # 5